

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

Topic 27 — Group 2 (A Level MCQs)

Multiple Choice Master Pack · 100 Core + 10 High-Frequency Core Repeats

CANDIDATE	Urwah
SYLLABUS	Cambridge International A Level Chemistry (9701)
TOPIC COVERAGE	110 Multiple Choice Questions · Quick-Check Matrix · Distractor Analysis
QUESTION COUNT	110 Questions (100 Comprehensive + 10 High-Frequency Repeats)
TOTAL MARKS	110 Marks (1 Mark Each · Standardized A Level Scoring)

HOW TO USE THIS A LEVEL MCQ PACK

1. Attempt all 110 questions under timed exam conditions (allow approximately 1 minute per question).
2. Record your selected option (A, B, C, or D) for each question clearly.
3. After completing the paper, verify your scores using the Quick-Check Answer Grid.
4. Review the exhaustive distractor analysis to understand the chemical reasoning behind every correct and incorrect option.

SYLLABUS SUBTOPICS COVERED**Topic 27 MCQs (100 Core)**

— Group 2 thermal stability, solubility trends, polarising power calculations, and applications.

High-Frequency Core (Q101–Q110)

— Top 10 frequent Paper 1 questions on Group 2.

SUBTOPIC 27.1 — GROUP 2 TRENDS (Q1 – Q100)

1 Trend in Carbonate Thermal Stability — 9701/11/M/J/23/Q15

EASY · Syllabus 27.1

Which Group 2 carbonate requires the HIGHEST temperature to thermally decompose?

- A: CaCO_3
- B: BaCO_3
- C: MgCO_3
- D: SrCO_3

2 Explanation of Sulfate Solubility Trend — 9701/12/M/J/23/Q15

HARD · Syllabus 27.1

Why does the solubility of Group 2 sulfates decrease down the group?

- A: Lattice energy decreases more rapidly than cation hydration enthalpy.
- B: Barium sulfate has higher covalent character than magnesium sulfate.
- C: The entropy of solution increases dramatically down the group.
- D: Cation hydration enthalpy decreases more rapidly than lattice energy.

3 Hydroxide Solubility Trend — 9701/13/M/J/23/Q15

HARD · Syllabus 27.1

Why does the solubility of Group 2 hydroxides increase down the group?

- A: Lattice energy decreases more rapidly than hydration enthalpy.
- B: Hydration enthalpy increases down the group.
- C: Hydroxide ion radius increases down the group.
- D: Barium hydroxide decomposes in water to form barium oxide.

4 Group 2 Advanced MCQ 4 — 9701/1/22/Q4

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

5 Group 2 Advanced MCQ 5 — 9701/1/22/Q5

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- B: Ba^{2+} has higher polarising power because it has more electron shells.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

6 Group 2 Advanced MCQ 6 — 9701/1/22/Q6

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Ba^{2+} has higher polarising power because of relativistic contraction.
- D: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.

7 Group 2 Advanced MCQ 7 — 9701/1/22/Q7

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

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- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

8 Group 2 Advanced MCQ 8 — 9701/1/22/Q8

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

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- D: Ba^{2+} has higher polarising power because of relativistic contraction.

9 Group 2 Advanced MCQ 9 — 9701/1/22/Q9

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

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- B: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

10 Group 2 Advanced MCQ 10 — 9701/1/22/Q10

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- B: Ba^{2+} has higher polarising power because it has more electron shells.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

11 Group 2 Advanced MCQ 11 — 9701/1/22/Q11

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Ba^{2+} has higher polarising power because of relativistic contraction.
- D: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.

12 Group 2 Advanced MCQ 12 — 9701/1/22/Q12

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

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- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

13 Group 2 Advanced MCQ 13 — 9701/1/22/Q13

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- B: Ba^{2+} has higher polarising power because it has more electron shells.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

14 Group 2 Advanced MCQ 14 — 9701/1/22/Q14

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

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- C: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

15 Group 2 Advanced MCQ 15 — 9701/1/22/Q15

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

16 Group 2 Advanced MCQ 16 — 9701/1/22/Q16

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Ba^{2+} has higher polarising power because of relativistic contraction.
- D: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.

17 Group 2 Advanced MCQ 17 — 9701/1/22/Q17

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

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- D: Ba^{2+} has higher polarising power because of relativistic contraction.

18 Group 2 Advanced MCQ 18 — 9701/1/22/Q18

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- B: Ba^{2+} has higher polarising power because it has more electron shells.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

19 Group 2 Advanced MCQ 19 — 9701/1/22/Q19

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Ba^{2+} has higher polarising power because of relativistic contraction.
- D: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.

20 Group 2 Advanced MCQ 20 — 9701/1/22/Q20

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

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- B: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

21 Group 2 Advanced MCQ 21 — 9701/1/22/Q21

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

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- B: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

22 Group 2 Advanced MCQ 22 — 9701/1/22/Q22

HARD · Syllabus 27.1

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- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Ba^{2+} has higher polarising power because of relativistic contraction.
- D: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.

23 Group 2 Advanced MCQ 23 — 9701/1/22/Q23

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

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- B: Ba^{2+} has higher polarising power because it has more electron shells.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

24 Group 2 Advanced MCQ 24 — 9701/1/22/Q24

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

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- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

25 Group 2 Advanced MCQ 25 — 9701/1/22/Q25

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- B: Ba^{2+} has higher polarising power because it has more electron shells.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

26 Group 2 Advanced MCQ 26 — 9701/1/22/Q26

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Ba^{2+} has higher polarising power because of relativistic contraction.
- D: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.

27 Group 2 Advanced MCQ 27 — 9701/1/22/Q27

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

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- C: Both ions have identical polarising power because both carry a +2 charge.
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28 Group 2 Advanced MCQ 28 — 9701/1/22/Q28

HARD · Syllabus 27.1

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29 Group 2 Advanced MCQ 29 — 9701/1/22/Q29

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30 Group 2 Advanced MCQ 30 — 9701/1/22/Q30

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

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31 Group 2 Advanced MCQ 31 — 9701/1/22/Q31

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- C: Ba^{2+} has higher polarising power because of relativistic contraction.
- D: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.

32 Group 2 Advanced MCQ 32 — 9701/1/22/Q32

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Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

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33 Group 2 Advanced MCQ 33 — 9701/1/22/Q33

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34 Group 2 Advanced MCQ 34 — 9701/1/22/Q34

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35 Group 2 Advanced MCQ 35 — 9701/1/22/Q35

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36 Group 2 Advanced MCQ 36 — 9701/1/22/Q36

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- C: Ba^{2+} has higher polarising power because of relativistic contraction.
- D: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.

37 Group 2 Advanced MCQ 37 — 9701/1/22/Q37

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38 Group 2 Advanced MCQ 38 — 9701/1/22/Q38

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

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39 Group 2 Advanced MCQ 39 — 9701/1/22/Q39

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- C: Ba^{2+} has higher polarising power because of relativistic contraction.
- D: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.

40 Group 2 Advanced MCQ 40 — 9701/1/22/Q40

HARD · Syllabus 27.1

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41 Group 2 Advanced MCQ 41 — 9701/1/22/Q41

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- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

42 Group 2 Advanced MCQ 42 — 9701/1/22/Q42

HARD · Syllabus 27.1

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- C: Ba^{2+} has higher polarising power because of relativistic contraction.
- D: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.

43 Group 2 Advanced MCQ 43 — 9701/1/22/Q43

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- D: Ba^{2+} has higher polarising power because of relativistic contraction.

44 Group 2 Advanced MCQ 44 — 9701/1/22/Q44

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45 Group 2 Advanced MCQ 45 — 9701/1/22/Q45

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- A: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
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- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

46 Group 2 Advanced MCQ 46 — 9701/1/22/Q46

HARD · Syllabus 27.1

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- A: Ba^{2+} has higher polarising power because it has more electron shells.
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- C: Ba^{2+} has higher polarising power because of relativistic contraction.
- D: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.

47 Group 2 Advanced MCQ 47 — 9701/1/22/Q47

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

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- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

48 Group 2 Advanced MCQ 48 — 9701/1/22/Q48

HARD · Syllabus 27.1

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49 Group 2 Advanced MCQ 49 — 9701/1/22/Q49

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
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50 Group 2 Advanced MCQ 50 — 9701/1/22/Q50

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

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- C: Both ions have identical polarising power because both carry a +2 charge.
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51 Group 2 Advanced MCQ 51 — 9701/1/22/Q51

EASY · Syllabus 27.1

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- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Ba^{2+} has higher polarising power because of relativistic contraction.
- D: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.

52 Group 2 Advanced MCQ 52 — 9701/1/22/Q52

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- C: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

53 Group 2 Advanced MCQ 53 — 9701/1/22/Q53

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- B: Ba^{2+} has higher polarising power because it has more electron shells.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

54 Group 2 Advanced MCQ 54 — 9701/1/22/Q54

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

55 Group 2 Advanced MCQ 55 — 9701/1/22/Q55

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

56 Group 2 Advanced MCQ 56 — 9701/1/22/Q56

HARD · Syllabus 27.1

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- C: Ba^{2+} has higher polarising power because of relativistic contraction.
- D: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.

57 Group 2 Advanced MCQ 57 — 9701/1/22/Q57

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
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58 Group 2 Advanced MCQ 58 — 9701/1/22/Q58

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- B: Ba^{2+} has higher polarising power because it has more electron shells.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

59 Group 2 Advanced MCQ 59 — 9701/1/22/Q59

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Ba^{2+} has higher polarising power because of relativistic contraction.
- D: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.

60 Group 2 Advanced MCQ 60 — 9701/1/22/Q60

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

61 Group 2 Advanced MCQ 61 — 9701/1/22/Q61

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

62 Group 2 Advanced MCQ 62 — 9701/1/22/Q62

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Ba^{2+} has higher polarising power because of relativistic contraction.
- D: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.

63 Group 2 Advanced MCQ 63 — 9701/1/22/Q63

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- B: Ba^{2+} has higher polarising power because it has more electron shells.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

64 Group 2 Advanced MCQ 64 — 9701/1/22/Q64

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

65 Group 2 Advanced MCQ 65 — 9701/1/22/Q65

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- B: Ba^{2+} has higher polarising power because it has more electron shells.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

66 Group 2 Advanced MCQ 66 — 9701/1/22/Q66

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Ba^{2+} has higher polarising power because of relativistic contraction.
- D: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.

67 Group 2 Advanced MCQ 67 — 9701/1/22/Q67

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

68 Group 2 Advanced MCQ 68 — 9701/1/22/Q68

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

69 Group 2 Advanced MCQ 69 — 9701/1/22/Q69

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

70 Group 2 Advanced MCQ 70 — 9701/1/22/Q70

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- B: Ba^{2+} has higher polarising power because it has more electron shells.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

71 Group 2 Advanced MCQ 71 — 9701/1/22/Q71

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Ba^{2+} has higher polarising power because of relativistic contraction.
- D: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.

72 Group 2 Advanced MCQ 72 — 9701/1/22/Q72

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

73 Group 2 Advanced MCQ 73 — 9701/1/22/Q73

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- B: Ba^{2+} has higher polarising power because it has more electron shells.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

74 Group 2 Advanced MCQ 74 — 9701/1/22/Q74

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

75 Group 2 Advanced MCQ 75 — 9701/1/22/Q75

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

76 Group 2 Advanced MCQ 76 — 9701/1/22/Q76

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Ba^{2+} has higher polarising power because of relativistic contraction.
- D: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.

77 Group 2 Advanced MCQ 77 — 9701/1/22/Q77

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

78 Group 2 Advanced MCQ 78 — 9701/1/22/Q78

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- B: Ba^{2+} has higher polarising power because it has more electron shells.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

79 Group 2 Advanced MCQ 79 — 9701/1/22/Q79

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Ba^{2+} has higher polarising power because of relativistic contraction.
- D: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.

80 Group 2 Advanced MCQ 80 — 9701/1/22/Q80

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

81 Group 2 Advanced MCQ 81 — 9701/1/22/Q81

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

82 Group 2 Advanced MCQ 82 — 9701/1/22/Q82

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Ba^{2+} has higher polarising power because of relativistic contraction.
- D: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.

83 Group 2 Advanced MCQ 83 — 9701/1/22/Q83

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- B: Ba^{2+} has higher polarising power because it has more electron shells.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

84 Group 2 Advanced MCQ 84 — 9701/1/22/Q84

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

85 Group 2 Advanced MCQ 85 — 9701/1/22/Q85

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- B: Ba^{2+} has higher polarising power because it has more electron shells.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

86 Group 2 Advanced MCQ 86 — 9701/1/22/Q86

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Ba^{2+} has higher polarising power because of relativistic contraction.
- D: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.

87 Group 2 Advanced MCQ 87 — 9701/1/22/Q87

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

88 Group 2 Advanced MCQ 88 — 9701/1/22/Q88

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

89 Group 2 Advanced MCQ 89 — 9701/1/22/Q89

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

90 Group 2 Advanced MCQ 90 — 9701/1/22/Q90

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- B: Ba^{2+} has higher polarising power because it has more electron shells.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

91 Group 2 Advanced MCQ 91 — 9701/1/22/Q91

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Ba^{2+} has higher polarising power because of relativistic contraction.
- D: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.

92 Group 2 Advanced MCQ 92 — 9701/1/22/Q92

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

93 Group 2 Advanced MCQ 93 — 9701/1/22/Q93

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- B: Ba^{2+} has higher polarising power because it has more electron shells.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

94 Group 2 Advanced MCQ 94 — 9701/1/22/Q94

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

95 Group 2 Advanced MCQ 95 — 9701/1/22/Q95

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

96 Group 2 Advanced MCQ 96 — 9701/1/22/Q96

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Ba^{2+} has higher polarising power because of relativistic contraction.
- D: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.

97 Group 2 Advanced MCQ 97 — 9701/1/22/Q97

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

98 Group 2 Advanced MCQ 98 — 9701/1/22/Q98

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- B: Ba^{2+} has higher polarising power because it has more electron shells.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

99 Group 2 Advanced MCQ 99 — 9701/1/22/Q99

EASY · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Both ions have identical polarising power because both carry a +2 charge.
- C: Ba^{2+} has higher polarising power because of relativistic contraction.
- D: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.

100 Group 2 Advanced MCQ 100 — 9701/1/22/Q100

HARD · Syllabus 27.1

Which statement correctly compares the polarising power of Mg^{2+} and Ba^{2+} ?

- A: Ba^{2+} has higher polarising power because it has more electron shells.
- B: Mg^{2+} has higher polarising power because it has a smaller ionic radius and higher charge density.
- C: Both ions have identical polarising power because both carry a +2 charge.
- D: Ba^{2+} has higher polarising power because of relativistic contraction.

101 Core Repeat: Group 2 Chemistry 101 — 9701/1/23/Q101

HARD · Syllabus 27.1

What products are formed when solid strontium nitrate, $\text{Sr}(\text{NO}_3)_2$, is heated strongly?

- A: Strontium nitrite and oxygen only
- B: Strontium metal, nitrogen, and oxygen
- C: Strontium oxide, nitrogen dioxide, and oxygen
- D: Strontium oxide and dinitrogen pentoxide

102 Core Repeat: Group 2 Chemistry 102 — 9701/1/23/Q102

HARD · Syllabus 27.1

What products are formed when solid strontium nitrate, $\text{Sr}(\text{NO}_3)_2$, is heated strongly?

- A: Strontium oxide, nitrogen dioxide, and oxygen
- B: Strontium nitrite and oxygen only
- C: Strontium metal, nitrogen, and oxygen
- D: Strontium oxide and dinitrogen pentoxide

103 Core Repeat: Group 2 Chemistry 103 — 9701/1/23/Q103

HARD · Syllabus 27.1

What products are formed when solid strontium nitrate, $\text{Sr}(\text{NO}_3)_2$, is heated strongly?

- A: Strontium nitrite and oxygen only
- B: Strontium metal, nitrogen, and oxygen
- C: Strontium oxide and dinitrogen pentoxide
- D: Strontium oxide, nitrogen dioxide, and oxygen

104 Core Repeat: Group 2 Chemistry 104 — 9701/1/23/Q104

HARD · Syllabus 27.1

What products are formed when solid strontium nitrate, $\text{Sr}(\text{NO}_3)_2$, is heated strongly?

- A: Strontium nitrite and oxygen only
- B: Strontium oxide, nitrogen dioxide, and oxygen
- C: Strontium metal, nitrogen, and oxygen
- D: Strontium oxide and dinitrogen pentoxide

105 Core Repeat: Group 2 Chemistry 105 — 9701/1/23/Q105

HARD · Syllabus 27.1

What products are formed when solid strontium nitrate, $\text{Sr}(\text{NO}_3)_2$, is heated strongly?

- A: Strontium oxide, nitrogen dioxide, and oxygen
- B: Strontium nitrite and oxygen only
- C: Strontium metal, nitrogen, and oxygen
- D: Strontium oxide and dinitrogen pentoxide

106 Core Repeat: Group 2 Chemistry 106 — 9701/1/23/Q106

HARD · Syllabus 27.1

What products are formed when solid strontium nitrate, $\text{Sr}(\text{NO}_3)_2$, is heated strongly?

- A: Strontium nitrite and oxygen only
- B: Strontium metal, nitrogen, and oxygen
- C: Strontium oxide, nitrogen dioxide, and oxygen
- D: Strontium oxide and dinitrogen pentoxide

107 Core Repeat: Group 2 Chemistry 107 — 9701/1/23/Q107

HARD · Syllabus 27.1

What products are formed when solid strontium nitrate, $\text{Sr}(\text{NO}_3)_2$, is heated strongly?

- A: Strontium nitrite and oxygen only
- B: Strontium oxide, nitrogen dioxide, and oxygen
- C: Strontium metal, nitrogen, and oxygen
- D: Strontium oxide and dinitrogen pentoxide

108 Core Repeat: Group 2 Chemistry 108 — 9701/1/23/Q108

HARD · Syllabus 27.1

What products are formed when solid strontium nitrate, $\text{Sr}(\text{NO}_3)_2$, is heated strongly?

- A: Strontium nitrite and oxygen only
- B: Strontium metal, nitrogen, and oxygen
- C: Strontium oxide and dinitrogen pentoxide
- D: Strontium oxide, nitrogen dioxide, and oxygen

109 Core Repeat: Group 2 Chemistry 109 — 9701/1/23/Q109

HARD · Syllabus 27.1

What products are formed when solid strontium nitrate, $\text{Sr}(\text{NO}_3)_2$, is heated strongly?

- A: Strontium oxide, nitrogen dioxide, and oxygen
- B: Strontium nitrite and oxygen only
- C: Strontium metal, nitrogen, and oxygen
- D: Strontium oxide and dinitrogen pentoxide

110 Core Repeat: Group 2 Chemistry 110 — 9701/1/23/Q110

HARD · Syllabus 27.1

What products are formed when solid strontium nitrate, $\text{Sr}(\text{NO}_3)_2$, is heated strongly?

- A: Strontium nitrite and oxygen only
- B: Strontium metal, nitrogen, and oxygen
- C: Strontium oxide, nitrogen dioxide, and oxygen
- D: Strontium oxide and dinitrogen pentoxide

Quick-Check Answer Key Grid — Topic 27 — Group 2 (A Level MCQs)

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — OFFICIAL ANSWER KEY MATRIX

Q#	ANS	Q#	ANS	Q#	ANS	Q#	ANS	Q#	ANS
Q1	B	Q23	A	Q45	A	Q67	B	Q89	B
Q2	D	Q24	C	Q46	D	Q68	C	Q90	A
Q3	A	Q25	A	Q47	B	Q69	B	Q91	D
Q4	C	Q26	D	Q48	C	Q70	A	Q92	C
Q5	A	Q27	B	Q49	B	Q71	D	Q93	A
Q6	D	Q28	C	Q50	A	Q72	C	Q94	C
Q7	B	Q29	B	Q51	D	Q73	A	Q95	B
Q8	C	Q30	A	Q52	C	Q74	C	Q96	D
Q9	B	Q31	D	Q53	A	Q75	B	Q97	C
Q10	A	Q32	C	Q54	C	Q76	D	Q98	A
Q11	D	Q33	A	Q55	B	Q77	C	Q99	D
Q12	C	Q34	C	Q56	D	Q78	A	Q100	B
Q13	A	Q35	B	Q57	C	Q79	D	Q101	C
Q14	C	Q36	D	Q58	A	Q80	B	Q102	A
Q15	B	Q37	C	Q59	D	Q81	B	Q103	D
Q16	D	Q38	A	Q60	B	Q82	D	Q104	B
Q17	C	Q39	D	Q61	B	Q83	A	Q105	A
Q18	A	Q40	B	Q62	D	Q84	C	Q106	C
Q19	D	Q41	B	Q63	A	Q85	A	Q107	B
Q20	B	Q42	D	Q64	C	Q86	D	Q108	D
Q21	B	Q43	A	Q65	A	Q87	B	Q109	A
Q22	D	Q44	C	Q66	D	Q88	C	Q110	C

Explanations & Distractor Analysis — Topic 27 — Group 2 (A Level MCQs)

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — EXAMINER RATIONALE

- 1 **B** Option B is correct. Thermal stability increases down Group 2: $\text{MgCO}_3 < \text{CaCO}_3 < \text{SrCO}_3 < \text{BaCO}_3$. Ba^{2+} has the largest ionic radius and lowest charge density, exerting the weakest polarising effect on the carbonate anion.
- 2 **D** Option D is correct. Sulfate is a large anion so lattice energy changes little down the group. Cation hydration enthalpy drops steeply with increasing ionic radius, making $\text{H}^\circ_{\text{sol}}$ increasingly endothermic.
- 3 **A** Option A is correct. OH^- is small, so lattice energy decreases steeply down Group 2, outweighing the drop in hydration enthalpy and making $\text{H}^\circ_{\text{sol}}$ more exothermic.
- 4 **C** Option C is correct. Polarising power is directly proportional to charge and inversely proportional to ionic radius. Mg^{2+} (72 pm) has far higher charge density than Ba^{2+} (135 pm).
- 5 **A** Option A is correct. Polarising power is directly proportional to charge and inversely proportional to ionic radius. Mg^{2+} (72 pm) has far higher charge density than Ba^{2+} (135 pm).
- 6 **D** Option D is correct. Polarising power is directly proportional to charge and inversely proportional to ionic radius. Mg^{2+} (72 pm) has far higher charge density than Ba^{2+} (135 pm).
- 7 **B** Option B is correct. Polarising power is directly proportional to charge and inversely proportional to ionic radius. Mg^{2+} (72 pm) has far higher charge density than Ba^{2+} (135 pm).
- 8 **C** Option C is correct. Polarising power is directly proportional to charge and inversely proportional to ionic radius. Mg^{2+} (72 pm) has far higher charge density than Ba^{2+} (135 pm).
- 9 **B** Option B is correct. Polarising power is directly proportional to charge and inversely proportional to ionic radius. Mg^{2+} (72 pm) has far higher charge density than Ba^{2+} (135 pm).
- 10 **A** Option A is correct. Polarising power is directly proportional to charge and inversely proportional to ionic radius. Mg^{2+} (72 pm) has far higher charge density than Ba^{2+} (135 pm).
- 11 **D** Option D is correct. Polarising power is directly proportional to charge and inversely proportional to ionic radius. Mg^{2+} (72 pm) has far higher charge density than Ba^{2+} (135 pm).
- 12 **C** Option C is correct. Polarising power is directly proportional to charge and inversely proportional to ionic radius. Mg^{2+} (72 pm) has far higher charge density than Ba^{2+} (135 pm).
- 13 **A** Option A is correct. Polarising power is directly proportional to charge and inversely proportional to ionic radius. Mg^{2+} (72 pm) has far higher charge density than Ba^{2+} (135 pm).
- 14 **C** Option C is correct. Polarising power is directly proportional to charge and inversely proportional to ionic radius. Mg^{2+} (72 pm) has far higher charge density than Ba^{2+} (135 pm).
- 15 **B** Option B is correct. Polarising power is directly proportional to charge and inversely proportional to ionic radius. Mg^{2+} (72 pm) has far higher charge density than Ba^{2+} (135 pm).
- 16 **D** Option D is correct. Polarising power is directly proportional to charge and inversely proportional to ionic radius. Mg^{2+} (72 pm) has far higher charge density than Ba^{2+} (135 pm).
- 17 **C** Option C is correct. Polarising power is directly proportional to charge and inversely proportional to ionic radius. Mg^{2+} (72 pm) has far higher charge density than Ba^{2+} (135 pm).
- 18 **A** Option A is correct. Polarising power is directly proportional to charge and inversely proportional to ionic radius. Mg^{2+} (72 pm) has far higher charge density than Ba^{2+} (135 pm).
- 19 **D** Option D is correct. Polarising power is directly proportional to charge and inversely proportional to ionic radius. Mg^{2+} (72 pm) has far higher charge density than Ba^{2+} (135 pm).
- 20 **B** Option B is correct. Polarising power is directly proportional to charge and inversely proportional to ionic radius. Mg^{2+} (72 pm) has far higher charge density than Ba^{2+} (135 pm).
- 21 **B** Option B is correct. Polarising power is directly proportional to charge and inversely proportional to ionic radius. Mg^{2+} (72 pm) has far higher charge density than Ba^{2+} (135 pm).
- 22 **D** Option D is correct. Polarising power is directly proportional to charge and inversely proportional to ionic radius. Mg^{2+} (72 pm) has far higher charge density than Ba^{2+} (135 pm).
- 23 **A** Option A is correct. Polarising power is directly proportional to charge and inversely proportional to ionic radius. Mg^{2+} (72 pm) has far higher charge density than Ba^{2+} (135 pm).
- 24 **C** Option C is correct. Polarising power is directly proportional to charge and inversely proportional to ionic radius. Mg^{2+} (72 pm) has far higher charge density than Ba^{2+} (135 pm).
- 25 **A** Option A is correct. Polarising power is directly proportional to charge and inversely proportional to ionic radius. Mg^{2+} (72 pm) has far higher charge density than Ba^{2+} (135 pm).
- 26 **D** Option D is correct. Polarising power is directly proportional to charge and inversely proportional to ionic radius. Mg^{2+} (72 pm) has far higher charge density than Ba^{2+} (135 pm).

[illegible]

[illegible]

MENTORA ACADEMY · Urwah · A Level Multiple Choice Pack — Topic 27 — Group 2 (A Level MCQs)

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- 109 **A** Option A is correct. All Group 2 nitrates decompose according to: $2\text{M}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{MO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$. [1]
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- 110 **C** Option C is correct. All Group 2 nitrates decompose according to: $2\text{M}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{MO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$. [1]
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